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Euclidean Geometry In Mathematical Olympiads

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Problem 1.38


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277546

1607 posts

Mar 8, 2018, 10:24 PM • 5 👍

#1

Problem 1.38

In the cyclic quadrilateral $ABCD$, let I_1 and I_2 denote the incenters of $\triangle ABC$ and DBC , respectively. Prove that I_1I_2BC is cyclic.

This post has been edited 1 time. Last edited by 277546, Mar 8, 2018, 10:24 PM

277546

1607 posts

Mar 8, 2018, 10:25 PM • 7 👍

#2

Solution

By The Incenter/Excenter Lemma, I_2BC has a circumcircle with center L , where L is the midpoint of arc BC . Similarly, I_1BC does as well. Thus, I_1I_2BC is cyclic.

277546

1607 posts

Mar 8, 2018, 10:26 PM • 4 👍

#3

Wait, do we have to address the case in which B is opposite A as well? Because the hints seem to refer to that.

ythomashu

6256 posts

Mar 8, 2018, 10:36 PM • 4 👍

#4

I don't see how that changes anything

277546

1607 posts

Mar 8, 2018, 10:46 PM • 4 👍

#5

Well L wouldn't be the midpoint of the same arc, it would be of a different one (i think)

WizardMath

2487 posts

Mar 9, 2018, 12:50 AM • 8 👍

#7

The fact that $ABCD$ is given in this order implies that the only configuration you need to take care of is the one you used to prove. The other configuration doesn't have the problem statement true.

277546

1607 posts

Mar 9, 2018, 12:58 AM • 4 👍

#8

Huh. Thats what I figured, so I was quite confused with the hints. Maybe it's a typo?

HardyRamanuj...

71 posts

Jun 17, 2018, 5:03 PM • 3 👍

#9

So i just solved this problem but I have solved the problem for the case in which A and B are opposite (proved $\angle I_2CI_1 = \angle I_2BI_1$ (simple angle chase))and proved that the statement is wrong if they are adjacent using the fact that $\angle BIC = 90 + \angle A/2$. But after seeing this , I am really confused.

Arthur.

133 posts

Jun 17, 2018, 5:31 PM • 2 👍

#10



HardyRamanujan wrote:

So i just solved this problem but I have solved the problem for the case in which A and B are opposite (proved $\angle I_2CI_1 = \angle I_2BI_1$ (simple angle chase))and proved that the statement is wrong if they are adjacent using the fact that $\angle BIC = 90 + \angle A/2$. But after seeing this , I am really confused.



WizardMath wrote:

The fact that $ABCD$ is given in this order implies that the only configuration you need to take care of is the one you used to prove. The other configuration doesn't

have the problem statement true.

@HardyRamunjan does that address your issue?

HardyRamanuj...

71 posts

Jun 17, 2018, 6:11 PM • 3 👍

#11

I mean that my solution (to which the hints apply) is the opposite of realquarterb's solution (which no one has claimed incorrect) so 1 of them is wrong.

dexhunt

2 posts

Jun 21, 2018, 6:19 AM • 3 👍

#12

@realquarterb Is it nessasary to write 'arc BC opposite A and D ' instead of just 'arc BC '

This post has been edited 3 times. Last edited by dexhunt, Jun 21, 2018, 6:22 AM
Reason: Trivial mistake

277546

1607 posts

Jun 22, 2018, 12:16 AM • 3 👍

#13



dexhunt wrote: